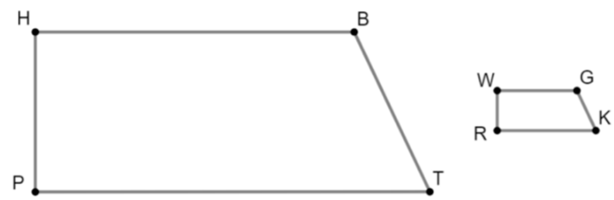
Geometry Name____________________
Unit 6 Date_____________________
Lesson 13 Homework Period____________________

Sheena, a party planner, is designing a banner for a birthday party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.



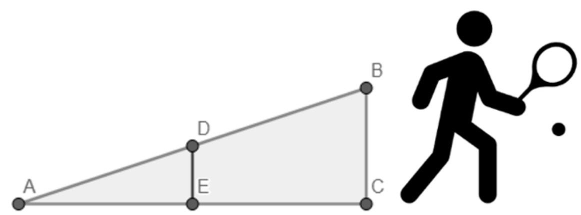
Sheena knows that $PT = 55$ inches (in), $HP = 3x + 2$ in, $WG = 8$ in, $WR = 0.5x + 1$ in, $BT = 5$ in, and $RK = 11$ in.

1. What scale factor will map $HBTP$ to $WGKR$?

2. What is the perimeter of $HBTP$, the main entrance banner?

3. What is the perimeter of $WGKR$, the photo booth banner?

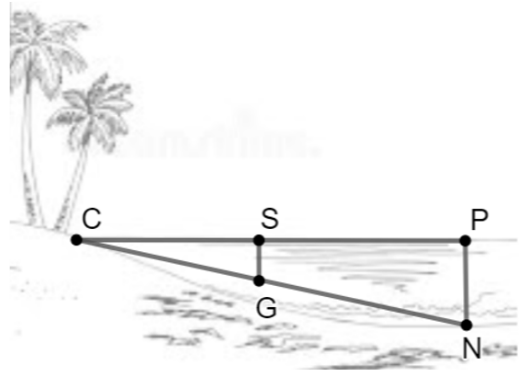
Venus is doing a tennis drill where she wants to hit the ball so that it will just pass over the net, $\overline{DE}$, and land 5 meters (m) away from the base of the net.



4. Given that $\triangle ADE \sim \triangle ABC$, $AE = 5$ m, $DE = 0.92$ m, and $EC = 10$ m, what is the height, represented by $\overline{BC}$, of the tennis racket?

5. What scale factor will map $\triangle ABC$ to $\triangle ADE$?

A beach front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SG}$ are the two main support posts for the dock.



$\triangle CSG \sim \triangle CPN$, $CS = 8$ feet (ft), $SG = (2x - 4)$ ft,

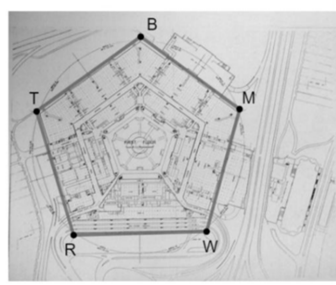
$PN = (5x + 8)$ ft, and $CP = 32$ ft.

6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?

7. What is the depth of the longer support post, represented by $\overline{PN}$?

8. What is the depth of the shorter support post, represented by $\overline{SG}$?

Cesar wants to create a scale model of the Pentagon for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{5}$ of the scale model Pentagon, $AKPSH$. $TB = \left(\frac{2}{3}x + 4\right)$ centimeters (cm) and $HA = (6x + 4)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?

10. What is the perimeter of the scale model Pentagon, $AKPSH$?